

A PRACTICAL SPECIFICATIONS FOR AUINN

Whitening transformation for $q(\mathbf{u}_d^l)$. The proposed parametric posterior for each unit is given by the Gaussian $q(\mathbf{u}_d^l) = \mathcal{N}(\mathbf{u}_d^l | \mathbf{m}_d^l, \mathbf{S}_d^l)$. The GP prior on $\mathbf{u}_d^l$ is $p(\mathbf{u}_d^l) = \mathcal{N}(\mathbf{u}_d^l | \boldsymbol{\mu}_d^l, \mathbf{K}_d^l)$, with $\boldsymbol{\mu}_d^l = \mu_d^l(\mathbf{z}_d^l)$ and $\mathbf{K}_d^l = k_d^l(\mathbf{z}_d^l, \mathbf{z}_d^l)$. For numerical stability and to reduce the amount of operations, we use a white representation for $q(\mathbf{u}_d^l)$, as is common practice in (D)GPs (De G. Matthews et al., 2017; Salimbeni & Deisenroth, 2017). That is, we consider the variable $\mathbf{v}_d^l \sim \mathcal{N}(\tilde{\mathbf{m}}_d^l, \tilde{\mathbf{S}}_d^l)$, with $\mathbf{u}_d^l = \boldsymbol{\mu}_d^l + (\mathbf{K}_d^l)^{1/2} \mathbf{v}_d^l$. Specifically, in the code the variable $\tilde{\mathbf{m}}_d^l$ is denoted as `q_mu`, and $\tilde{\mathbf{S}}_d^l$ is represented through its Cholesky factorization $(\tilde{\mathbf{S}}_d^l)^{1/2}$, which is named `q_sqrt`.

Initialization of the variational parameters $\{\mathbf{m}_d^l\}$. These are the mean of the posterior distribution on the inducing points. Therefore, their value determines the *initialization of the activation function*. If the RBF kernel is used, $\{\mathbf{m}_d^l\}$ are initialized to the prior $\boldsymbol{\mu}_d^l = \mu_d^l(\mathbf{z}_d^l)$ (since we are using the aforementioned white representation, `q_mu` is initialized to zero). This is the most standard initialization in GP literature. For the TRI kernel, $\{\mathbf{m}_d^l\}$ are initialized according to the ReLu which TRI is inspired by, i.e. $\mathbf{m}_d^l = \text{ReLu}(\mathbf{z}_d^l)$.

Initialization of the variational parameters $\{\mathbf{S}_d^l\}$. The posterior distribution covariance matrices are initialized to the prior $\mathbf{K}_d^l = k_d^l(\mathbf{z}_d^l, \mathbf{z}_d^l)$ (that is, `q_sqrt` is initialized to the identity matrix). Following common practise for DGPs (Salimbeni & Deisenroth, 2017), the covariance matrices of inner layers are scaled by 10^{-5} .

Initialization of the weights. The Glorot uniform initializer (Glorot & Bengio, 2010), also called Xavier uniform initializer, is used for the weights. The biases are initialized to zero.

Initialization of the kernel hyperparameters. The kernels used (RBF and TRI) have two hyperparameters: the variance γ and the lengthscale ℓ . Both are always initialized to 1 (except for the lengthscale in the 1D example in Section 3.1, where ℓ is initialized to 0.1).

Initialization of the inducing points. In order to initialize $\mathbf{z}_d^l$, the N input data points are propagated through the network with the aforementioned initial weights, biases, and activation function. Then, in each layer and unit, $\mathbf{z}_d^l$ is initialized with a `linspace` between the minimum and maximum of the N values there (the minimum (resp. the maximum) is decreased (resp. increased) by 0.1 to strictly contain the interval of interest).

Initialization of the regression likelihood noise. In the regression problems, we use a Gaussian likelihood $p(y|f) = \mathcal{N}(y|f, \sigma^2)$. The standard deviation of the noise is initialized to $\sigma = 0.1$.

Mean function. We always use a zero mean function. Since data is normalized to have zero mean (and standard deviation equal to one), a zero mean function allows for reverting to the empirical mean for OOD data, as explained in the main text.

Optimizer and learning rate. Throughout the work, we use the Adam Optimizer (Kingma & Ba, 2014) with default parameters and learning rate of 0.001.

B EXPERIMENTAL DETAILS FOR THE EXPERIMENTS

All the experiments were run on a NVIDIA Tesla P100. In order to predict, all the methods utilize 100 test samples in all the experiments. Details for each section are provided below.

An illustrative example (Section 3.1 in the main text). All the methods use two layers (i.e. one hidden layer). The hidden layer has $D = 25$ units in all cases. BNN and fBNN use ReLu activations. The auNN methods use $M = 10$ inducing points in each unit (the rest of methods do not have such inducing points). The methods are trained during 5000 epochs with the whole dataset (no mini-batches). The dataset is synthetically generated to have two clusters of points around $x = \pm 1$. More specifically, 30 points are sampled uniformly in each interval $(x - 0.3, x + 0.3)$ for $x = \pm 1$, and the output is given by the sin function plus a Gaussian noise of standard deviation 0.1. We have also trained DGP and GP on this dataset, see Figure 14. Both methods use $M = 10$ inducing points, and are trained during 5000 epochs with the whole dataset (no mini-batches). DGP has one hidden layer with $D = 25$ units.

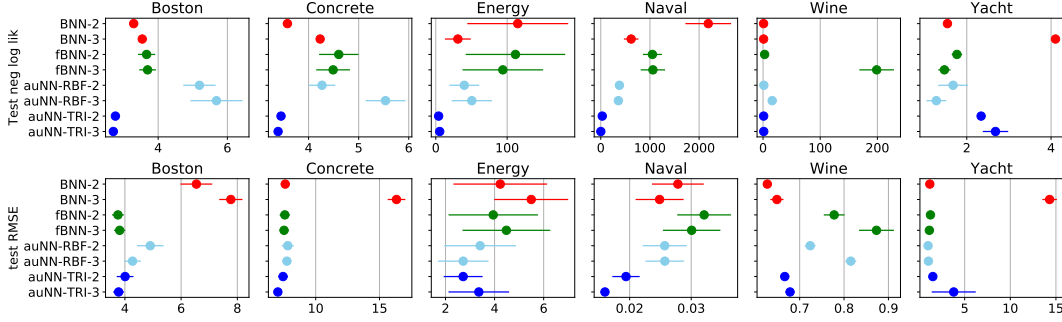


Figure 7: Performance of the compared methods in the gap splits for six UCI datasets. Mean and one standard error of NLL (upper row) and RMSE (lower row) are shown, the lower the better.

UCI regression datasets with gap (and standard) splits (Section 3.2 in the main text). The methods use $L = 2, 3$ layers. In all cases, the hidden layers have $D = 50$ units. BNN and fBNN use ReLu activations. The methods are trained during 10000 epochs, with a mini-batch size that depends on the size of the dataset. For those with fewer than 5000 instances (i.e. Boston, Concrete, Energy, Wine and Yacht), the mini-batch size is 500. For those with more than 5000 (i.e. Naval), the mini-batch size is 5000. Recall from the main text that each dataset has as many gap splits as dimensionality, with 2/3 for train and 1/3 for test. In the case of standard splits, each dataset uses 10 random 90%-10% train-test splits. Regarding the segment used in Figure 5, each extreme of the segment is a point from a different connected component of the training set. These are chosen so that the function is well-known in the extremes (but not along the segment, which crosses the gap). Namely, the extremes are chosen as the training points who have minimum average distance to the closest five points in its connected component.

Comparison with DGPs (Section 3.3 in the main text). Here, different values of depth L , number of inducing points M and number of hidden layers D are studied (see the main text). auNN is trained during 5000 epochs, with a mini-batch size of 5000 (20000 epochs are used for DGP, as proposed by the authors (Salimbeni & Deisenroth, 2017)). Each experiment is repeated on five random 90%-10% train-test splits. DGP uses a RBF kernel. The experimental details for DGP-add are the same as for DGP, with the only difference of the kernel. Namely, an additive kernel using RBF components is used for DGP-add.

Large scale experiments (Section 3.4 in the main text). Since we are dealing with classification datasets, a Robust-Max likelihood is used in all cases (Hernández-Lobato et al., 2011). The values of D and M are chosen following the conclusions from Section 3.3. That is, DGP needs large M (the largest $M = 100$ is used), but is less influenced by D (this is chosen as recommended by the authors (Salimbeni & Deisenroth, 2017): $D = \min(30, D^0)$, with D^0 the dimensionality of the input data). auNN needs large D (the largest $D = 50$ is used), but is less influenced by M (the intermediate value $M = 25$ is chosen). All the methods are trained during 100 epochs, with a mini-batch size of 5000. Three random train-test splits are used. In both datasets, 500000 instances are used for test (which leaves 10.5M and 4.5M training instances for HIGGS and SUSY, respectively).

C ADDITIONAL FIGURES AND TABLES

Finally, additional material is provided here. Every figure and table is referred from the main text.

Table 4: Test NLL for the gap splits of the six UCI datasets (mean and one standard error, the lower the better). Last column is the per-group (weight-space stochasticity vs activation-level stochasticity) average rank.

	Boston	Concrete	Energy	Naval	Wine	Yacht	Rank	Rank (group)
BNN-2	3.29±0.10	3.58±0.09	114.84±70.69	2186.30±464.32	0.96±0.01	1.54±0.09	3.92±0.79	4.83±0.32
BNN-3	3.54±0.03	4.23±0.04	30.91±19.97	618.44±147.99	0.98±0.02	4.10±0.03	4.98±0.70	
fBNN-2	3.67±0.25	4.60±0.39	111.65±69.68	1050.65±192.61	2.80±0.31	1.77±0.12	5.04±0.36	
fBNN-3	3.69±0.24	4.49±0.34	93.92±56.45	1060.54±247.21	198.76±30.24	1.47±0.15	5.36±0.50	
auNN-RBF-2	5.19±0.47	4.27±0.26	39.93±20.89	379.55±67.74	1.44±0.05	1.68±0.35	4.69±0.61	4.17±0.40
auNN-RBF-3	5.68±0.75	5.54±0.40	50.48±28.26	352.94±72.13	16.05±1.13	1.28±0.23	5.29±0.89	
auNN-TRI-2	2.77±0.06	3.45±0.06	3.99±1.14	30.47±5.54	1.06±0.03	2.34±0.03	3.25±0.57	
auNN-TRI-3	2.70±0.04	3.39±0.06	5.50±2.45	2.38±3.23	1.23±0.04	2.68±0.30	3.47±0.80	

Table 5: Test RMSE for the gap splits of the six UCI datasets (mean and one standard error, the lower the better). Last column is the per-group (weight-space stochasticity vs activation-level stochasticity) average rank.

	Boston	Concrete	Energy	Naval	Wine	Yacht	Rank	Rank (group)
BNN-2	6.54±0.56	7.62±0.35	4.23±1.91	0.03±0.00	0.63±0.01	1.18±0.11	4.09±0.67	4.91±0.37
BNN-3	7.77±0.40	16.33±0.67	5.27±1.41	0.02±0.00	0.64±0.01	14.31±0.76	6.15±0.91	
fBNN-2	3.75±0.21	7.58±0.41	3.95±1.82	0.03±0.00	0.78±0.02	1.25±0.08	4.70±0.54	
fBNN-3	3.81±0.20	7.52±0.36	4.48±1.79	0.03±0.00	0.87±0.04	1.13±0.12	4.71±0.53	
auNN-RBF-2	4.90±0.47	7.81±0.47	3.41±1.46	0.03±0.00	0.72±0.01	0.99±0.18	4.32±0.47	4.09±0.23
auNN-RBF-3	4.27±0.29	7.74±0.21	2.72±1.03	0.03±0.00	0.82±0.01	1.03±0.14	4.27±0.55	
auNN-TRI-2	4.01±0.30	7.44±0.38	2.72±0.79	0.02±0.00	0.67±0.01	1.51±0.20	3.90±0.33	
auNN-TRI-3	3.78±0.19	7.03±0.23	3.36±1.23	0.02±0.00	0.68±0.01	3.80±2.41	3.85±0.46	

Table 6: Test NLL for the standard splits of the six UCI datasets (mean and one standard error, the lower the better). Last column is the per-group (weight-space stochasticity vs activation-level stochasticity) average rank.

<i>test NLL</i>	Boston	Concrete	Energy	Naval	Wine	Yacht	Rank	Rank (group)
BNN-2	2.71±0.07	3.12±0.02	0.65±0.04	-5.38±0.59	0.99±0.02	1.01±0.07	3.78±0.41	4.5±0.39
BNN-3	3.62±0.05	4.24±0.01	0.80±0.03	-5.02±0.33	1.01±0.02	4.06±0.05	6.25±0.70	
fBNN-2	2.83±0.20	3.20±0.04	0.67±0.04	-6.17±0.02	1.55±0.08	0.77±0.02	4.13±0.57	
fBNN-3	2.75±0.14	3.13±0.05	0.65±0.03	-6.26±0.00	207.43±9.12	0.79±0.02	3.83±0.85	
auNN-RBF-2	3.38±0.30	3.14±0.05	0.63±0.03	-5.40±0.08	1.16±0.06	0.52±0.04	3.97±0.60	4.5±0.43
auNN-RBF-3	3.89±0.47	3.25±0.13	0.53±0.07	-5.69±0.03	8.98±1.51	0.54±0.03	4.42±0.85	
auNN-TRI-2	2.56±0.05	3.08±0.02	1.47±0.04	-4.81±0.07	0.96±0.03	2.25±0.02	4.78±0.92	
auNN-TRI-3	2.50±0.02	2.98±0.02	1.42±0.02	-3.43±0.32	1.10±0.07	2.26±0.01	4.83±1.01	

Table 7: Test RMSE for the standard splits of the six UCI datasets (mean and one standard error, the lower the better). Last column is the per-group (weight-space stochasticity vs activation-level stochasticity) average rank.

<i>test RMSE</i>	Boston	Concrete	Energy	Naval	Wine	Yacht	Rank	Rank (group)
BNN-2	3.47±0.34	5.49±0.13	0.45±0.02	0.00±0.00	0.65±0.01	0.68±0.08	4.70±0.48	4.59±0.41
BNN-3	8.89±0.45	16.71±0.20	0.51±0.02	0.00±0.00	0.67±0.02	13.49±0.94	6.50±0.64	
fBNN-2	2.80±0.21	5.34±0.13	0.47±0.02	0.00±0.00	0.70±0.02	0.33±0.04	3.70±0.61	
fBNN-3	2.74±0.16	5.07±0.12	0.46±0.02	0.00±0.00	0.83±0.02	0.36±0.04	3.45±0.88	
auNN-RBF-2	3.16±0.23	5.13±0.16	0.45±0.02	0.00±0.00	0.67±0.02	0.41±0.04	4.25±0.35	4.41±0.41
auNN-RBF-3	3.01±0.25	4.51±0.18	0.41±0.03	0.00±0.00	0.76±0.02	0.38±0.03	3.35±0.77	
auNN-TRI-2	3.00±0.26	5.21±0.10	0.72±0.02	0.00±0.00	0.62±0.02	1.15±0.14	5.40±0.80	
auNN-TRI-3	2.81±0.17	4.67±0.15	0.65±0.03	0.01±0.00	0.62±0.02	1.16±0.15	4.65±1.00	

Table 8: Standard error obtained by auNN and DGP in three splits of the large scale classification datasets HIGGS and SUSY.

			auNN						DGP		
	N	D	RBF-2	RBF-3	RBF-4	TRI-2	TRI-3	TRI-4	DGP-2	DGP-3	DGP-4
HIGGS	11M	28	0.0001	0.0006	0.0007	0.0003	0.0004	0.0008	0.0005	0.0009	0.0010
SUSY	5M	18	0.0004	0.0005	0.0005	0.0005	0.0005	0.0004	0.0005	0.0027	0.0035